

Analysis that has already been completed:

1. exploratory density estimation (KDE),
2. spatial discretization + hotspot visualization (9-zone cluster map),
3. inferential testing (chi-squared against randomness).
4. Moran's I/ Spatial Autocorrelation
5. Ripley's K Function/ Point Pattern Analysis (this included some Monte Carlo Testing I think)
6. Directional Statistics/ Spray Angle Modeling

Next steps:

1. Nearest Neighbor Analysis
2. Convex Hull / Spatial Footprint Metrics
3. Comparative Clustering Between Hitters (VERY Important)
4. Spatial Entropy (Excellent Addition)
5. Expected Outcome Layer (xBA / xwOBA)
6. Permutation / Monte Carlo Testing

Moran's I Test Analysis:

Moran's I statistic = 0.6365 with p value $< 2.2 * 10^{-16}$. Interpretation of Moran's I statistic is as follows: negative means dispersion, near 0 indicates randomness, and positive indicates spatial autocorrelation. Therefore, 0.64 indicates a very high spatial autocorrelation.

Nearby field regions tend to contain similar batted-ball frequencies rather than random distributions.

In baseball terms:

Bo Bichette's balls in play exhibit strong spatial clustering, meaning adjacent regions of the field tend to experience similarly high or similarly low concentrations of batted balls.

The p-value being essentially 0 means:

The observed clustering pattern is extremely unlikely to arise under complete spatial randomness.

Your chi-squared test showed:

- the distribution is not random *categorically*.

Moran's I now shows:

- the distribution is not random *spatially*.

That distinction matters a lot.

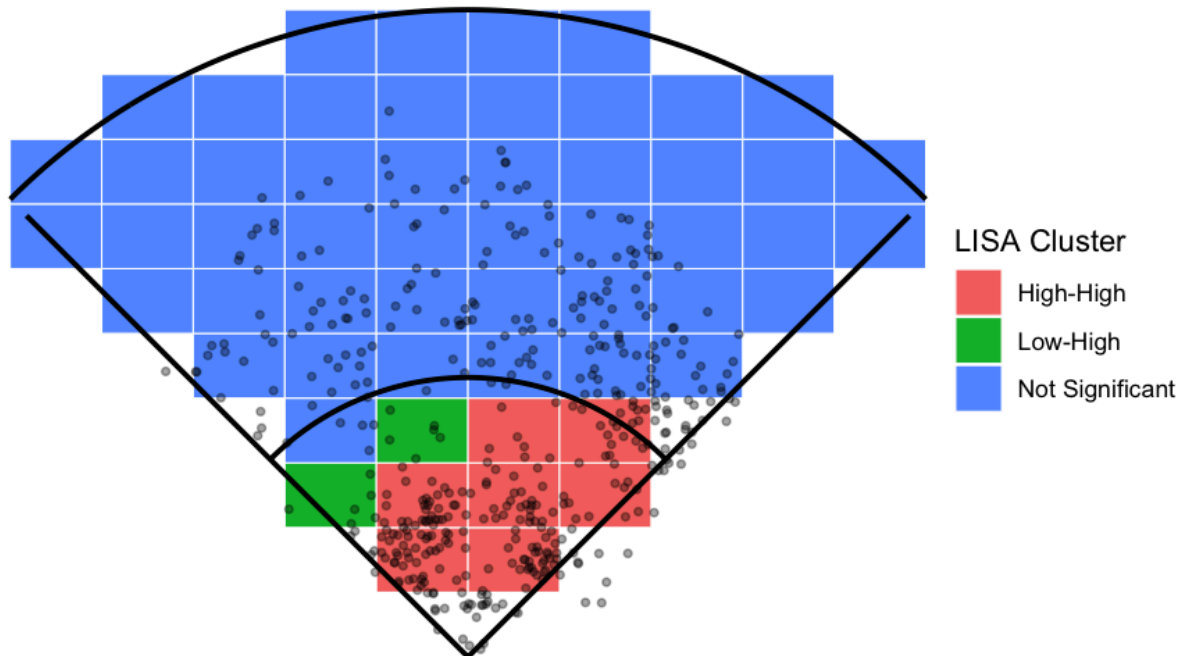
You can literally write:

While the chi-squared test established that Bo Bichette's batted-ball distribution deviates from randomness across predefined field zones, Moran's I demonstrates that these patterns also exhibit statistically significant spatial autocorrelation, meaning neighboring regions of the field tend to contain similar batted-ball frequencies.

LISA Analysis:

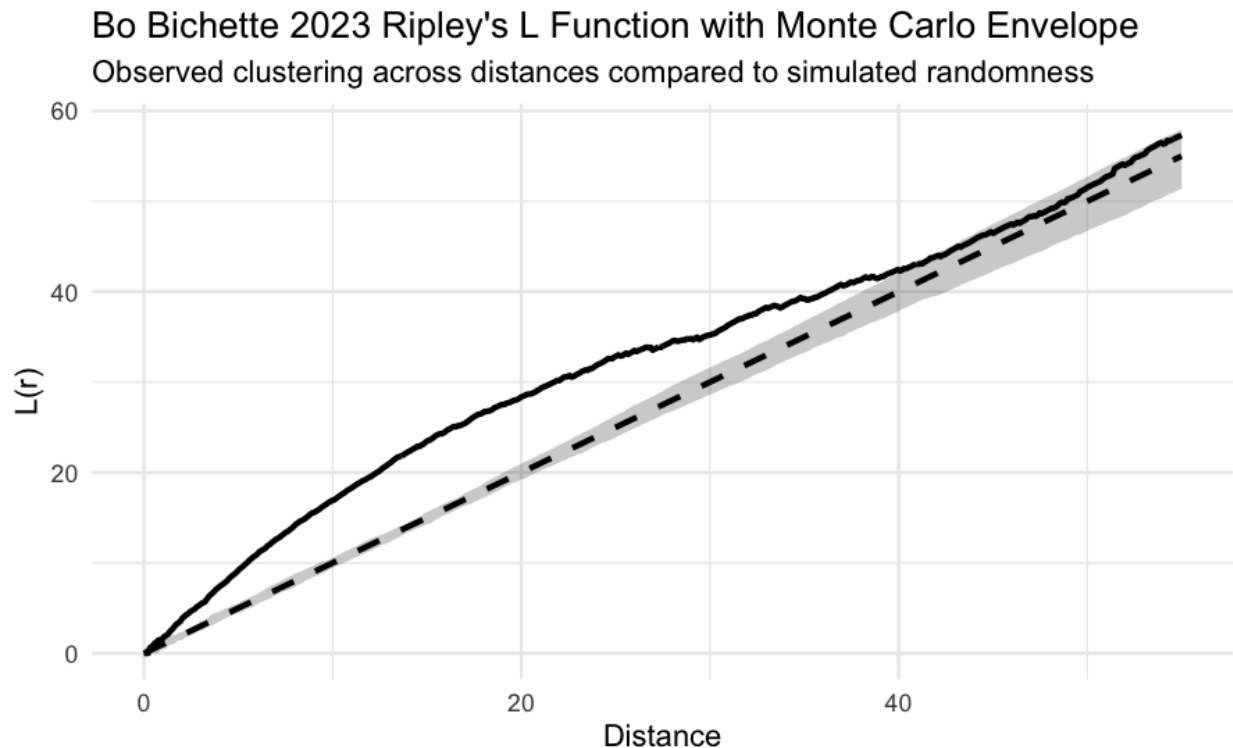
Bo Bichette 2023 Local Moran's I Cluster Map

Spatial autocorrelation of balls in play by field grid cell



The most statistically significant localized clusters occurred toward the opposite-field side of the infield.”

Ripley's K Function / Point Pattern Analysis:



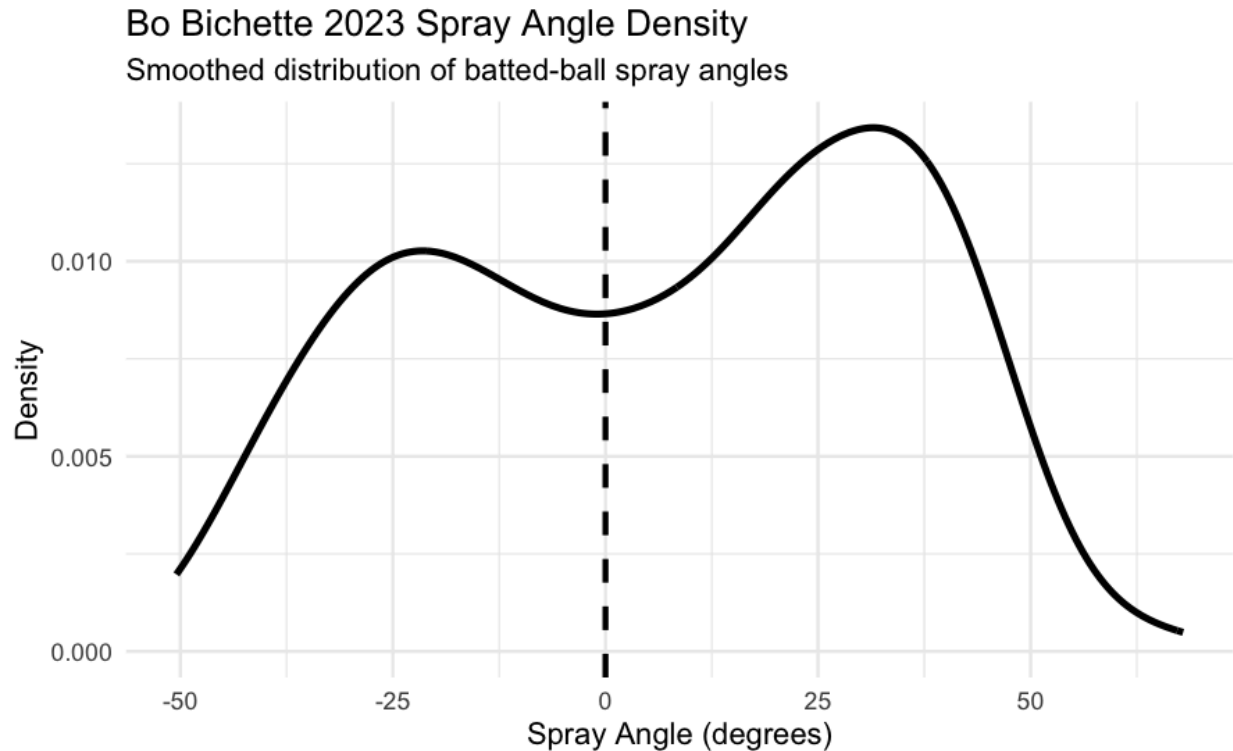
Null hypothesis: Complete Spatial Randomness (CSR)

Alternative hypothesis: the observed spatial point pattern deviates significantly from complete spatial randomness (CSR) at specified distances, indicating either clustering or dispersion.

- Ripley's L analysis demonstrates that Bo Bichette's batted ball locations exhibit statistically significant spatial clustering relative to complete spatial randomness across multiple spatial scales.
- The observed $L(r)$ curve exceeds both the theoretical CSR expectation and the Monte Carlo simulation envelope at shorter spatial distances, indicating strong localized clustering of balls in play.
- Clustering effects weaken as spatial scale increases, as the observed curve gradually converges toward the Monte Carlo envelope at larger distances. This suggests that Bichette's spray pattern is characterized by concentrated local hotspots rather than uniform clustering across the entire field.

- Because Ripley's L evaluates inter-point distances rather than distance from home plate directly, the analysis identifies multiscale spatial dependence rather than strictly infield or outfield clustering. However, the strongest clustering likely reflects concentrated contact within smaller spatial regions of the field.
- The Local Moran's I (LISA) cluster map complements the Ripley's L results by identifying where these localized clusters occur spatially, with significant high-high clusters appearing primarily on the opposite-field/right-side infield regions.
- Together, the Ripley's L and LISA analyses indicate that Bo Bichette's 2023 balls in play are not randomly distributed, but instead exhibit structured and repeatable spatial concentration patterns across the field.
- Unlike the 9-zone chi-squared analysis, Ripley's L evaluates clustering continuously across multiple spatial scales rather than within predefined grid boundaries, providing a more nuanced understanding of spatial dependence in batted ball locations.

Directional Statistics/ Spray Modeling:



- The spray angle analysis indicates that Bo Bichette's 2023 batted ball distribution exhibits a measurable opposite-field tendency rather than a perfectly centered spray profile. The mean spray angle was approximately 6.73 degrees, indicating that contact was shifted slightly toward the opposite-field side of the field.
- The histogram and density plots demonstrate that Bo Bichette's spray angle distribution is not symmetric around 0 degrees. Instead, the distribution contains a stronger concentration of positive spray angles, corresponding to opposite-field contact.
- The density plot suggests a multimodal spray profile, with one concentration of contact occurring on the pull side and a larger concentration occurring on the opposite-field side. This indicates that Bichette does not exhibit a uniformly distributed spray pattern, but instead tends to generate contact within specific directional ranges.
- Angle-based directional classification further supports this interpretation. Approximately 45.3% of modeled batted balls were classified as opposite field, compared to 28.4% pull side and 26.3% center field.
- These modeled spray-angle results are highly consistent with the previously calculated directional statistics (27% pull, 32.5% center, 40.5% opposite field), reinforcing the conclusion that Bichette demonstrates a clear opposite-field hitting tendency.

- The one-sample t-test strongly rejects the null hypothesis that the mean spray angle equals 0 degrees ($t = 5.16$, $p < 0.001$), providing statistical evidence that Bichette's spray distribution is significantly shifted away from a perfectly centered hitting profile.
- The directional spray-angle analysis complements the earlier KDE, LISA, and Ripley's L analyses by providing a continuous directional measure of contact tendency rather than relying solely on discrete spatial zones or clustering methods.
- Collectively, the results suggest that Bo Bichette's balls in play exhibit both statistically significant spatial clustering and a systematic directional preference toward the opposite-field side of the field.

One Sample Mean test Results:

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data:  bo_angles$spray_angle_deg

t = 5.1587, df = 456, p-value = 3.716e-07

alternative hypothesis: true mean is not equal to 0

95 percent confidence interval:

 4.163888 9.288531

sample estimates:

mean of x

 6.72621
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